



SIMATS Online Programming Lab

JAVA PROGRAMMING



KALAHSTHI RAMYA ✓
192311149RA

QUESTIONS

2 / 2 completed

Sahred

SAHRED

Write a Java program to demonstrate how to use synchronization to prevent multiple threads from accessing a shared resource simultaneously.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         SharedResource resource = new SharedResource();
4
5         // Creating multiple threads that will compete for the s
6         Worker t1 = new Worker(resource, "Thread-1");
7         Worker t2 = new Worker(resource, "Thread-2");
8         Worker t3 = new Worker(resource, "Thread-3");
9
10        t1.start();
11        t2.start();
12        t3.start();
13    }
14 }
15
16 class SharedResource {
17     private int counter = 0;
18
19     // The 'synchronized' keyword ensures only one thread can ex
20     public synchronized void increment(String threadName) {
21
```

OUTPUT

```
Thread-1 is entering the critical section.
Thread-1 incremented counter to: 1
Thread-1 is leaving the critical section.
Thread-1 is entering the critical section.
Thread-1 incremented counter to: 2
Thread-1 is leaving the critical section.
Thread-1 is entering the critical section.
Thread-1 incremented counter to: 3
Thread-1 is leaving the critical section.
Thread-3 is entering the critical section
```

INPUT

Your INPUT goes here!

© 2026 **SIMATS Online Lab**. All rights reserved.